

Task 2: Multiple Sequence Alignment

Tool name: Using **Clustal Omega** for Multiple Sequence Alignment (MSA).

Introduction: Multiple Sequence Alignment (MSA) is a computational approach used to align three or more biological sequences, such as DNA, RNA, or proteins, in order to identify regions of similarity. Clustal Omega is a new multiple sequence alignment program. These similarities often reveal important functional, structural, or evolutionary relationships among the sequences. Conserved regions in protein sequences typically indicate critical roles, such as active sites, binding domains, or motifs essential for protein stability and activity, while variable regions highlight species-specific adaptations, evolutionary divergence, or functional specialization.

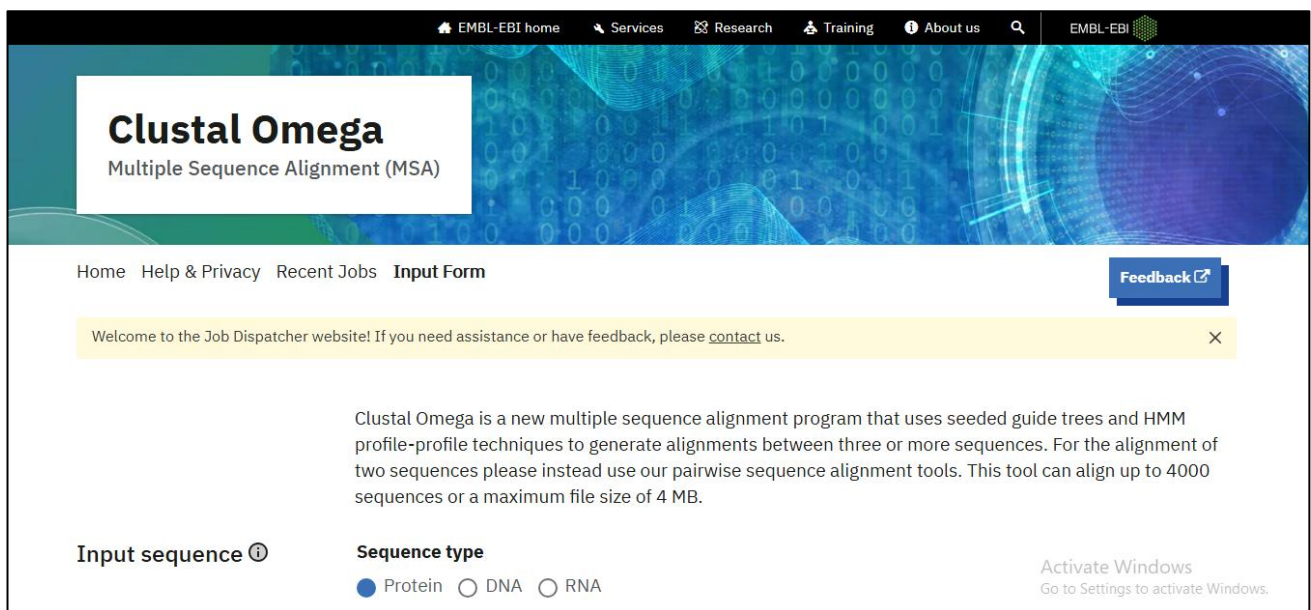


Fig 1: Clustal Omega homepage (<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>).



Fig 2: Conserved regions and motifs for MSA analysis.

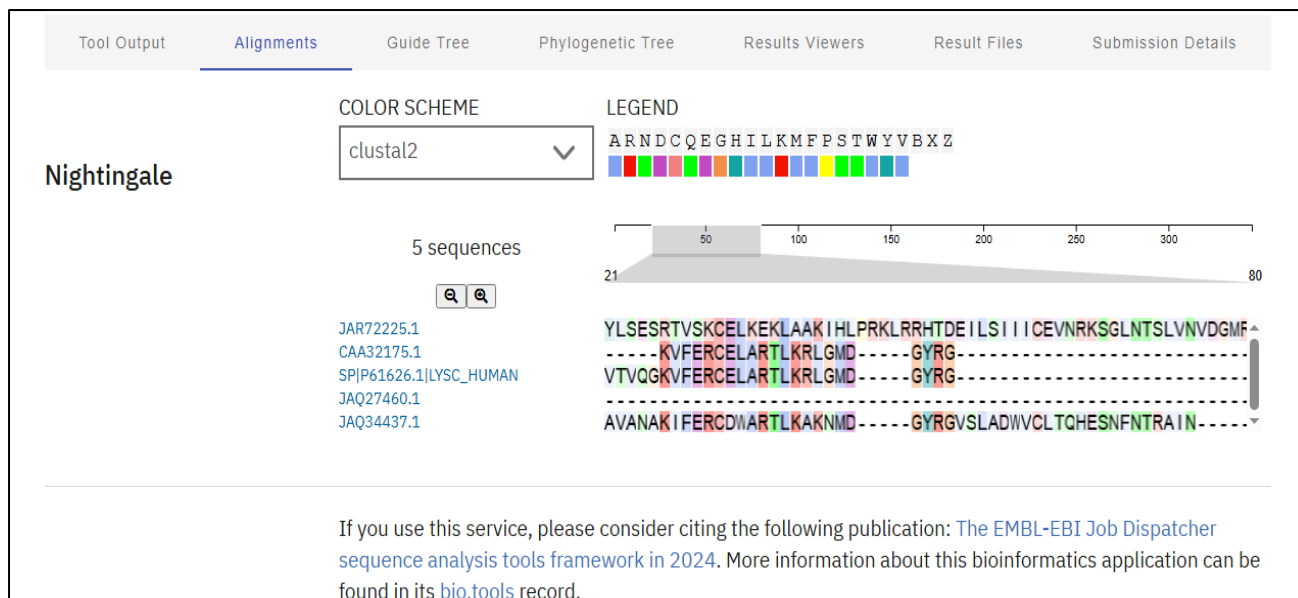


Fig 3: MSA alignments of selected organisms.

Result and Interpretation

The Multiple Sequence Alignment (MSA) revealed distinct conserved regions that indicate essential functional domains within the protein sequences. These conserved areas likely correspond to critical structural or catalytic roles, such as binding sites or motifs necessary for protein stability and activity. The Multiple Sequence Alignment of protein sequences from organisms belonging to the same family revealed several highly conserved regions and motifs. The high degree of sequence similarity indicates a close evolutionary relationship and suggests that these proteins perform similar biological functions across the analyzed species.